

The Build Fellowship

BY OPEN AVENUES



**Design of an Enclosure
for EV Battery**

By Zaid Khan

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Design & Manufacturing Strategy

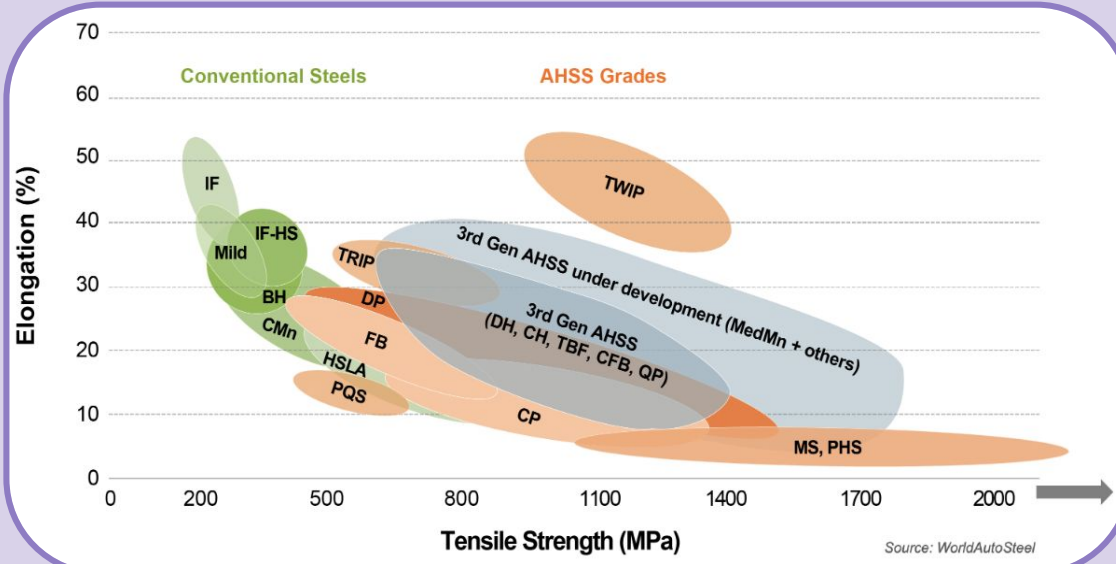
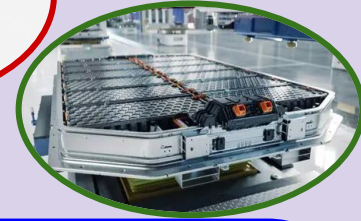
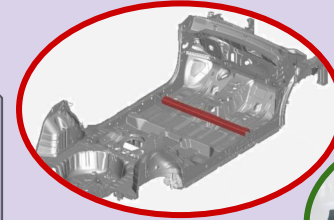


Design Requirements

- Lightweight and strong
- High capacity
- Cost effective
- Less parts
- Industry compliance

Applications of AHSS

- ❑ Automotive -
 - Crossmembers
 - Roof reinforcements
 - EV Enclosure, BIW

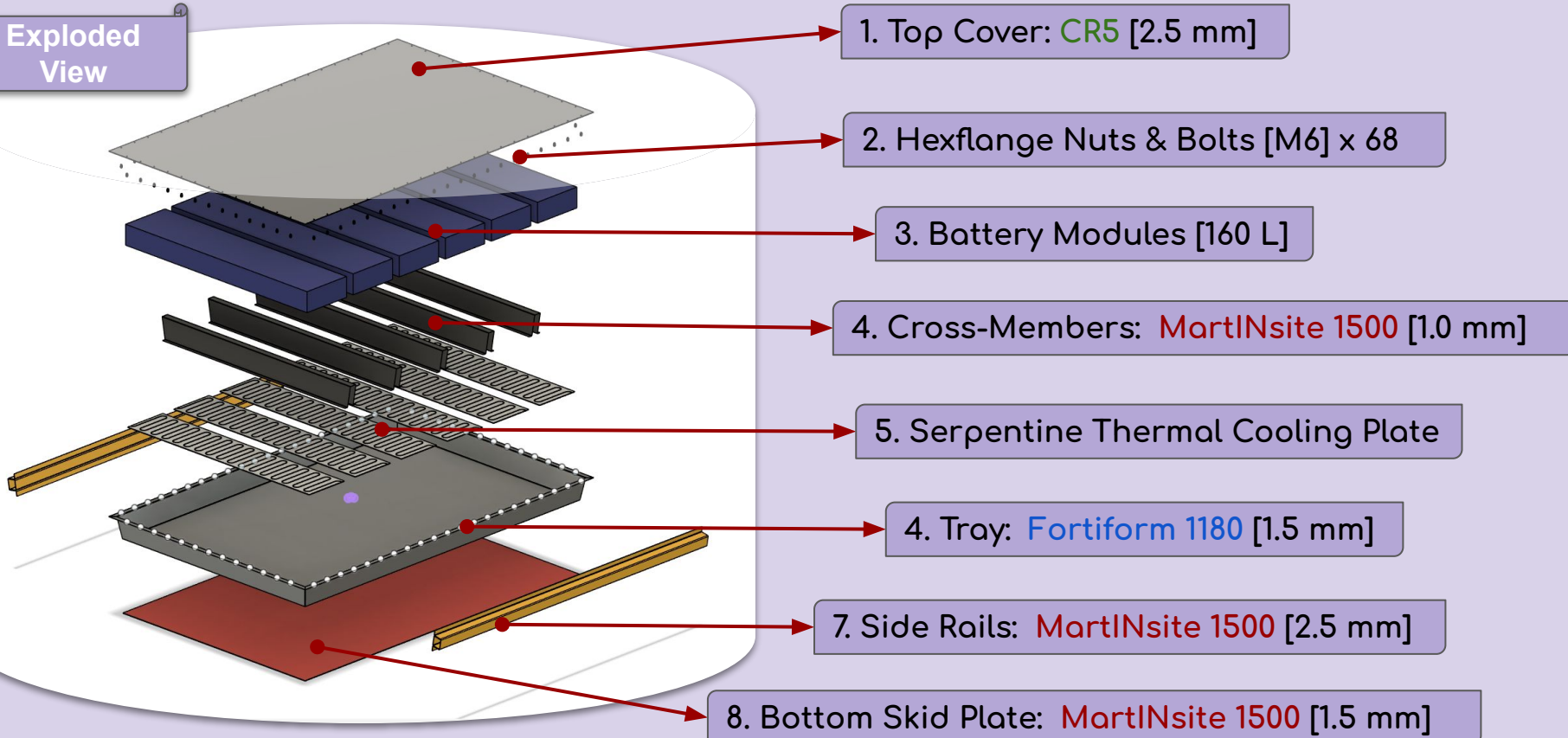


Parameter	Target Metric	Governing Standard	Design Justification
<i>Overall Envelope (L x W x H)</i>	2062 x 1343 x 128 (mm)	Ford Mach E	Maximizes LFP cell volume (160 L).
<i>Side Crush / Pole Impact</i>	Withstand 100 kN force (150mm rigid pole)	ECE R100 / FMVSS 305	Prevents side-intrusion into battery modules during a T-bone collision.
<i>Underfloor Intrusion</i>	Max 15 mm deflection at 35 kN load	Internal OEM Standard	Protects bottom shield from localized road debris strikes.
<i>First Modal Frequency</i>	> 35 Hz	Modal Analysis Target	Prevents the enclosure from hitting a resonant frequency.
<i>External Fire (Bonfire)</i>	70 seconds of direct flame without explosion	GB/T 31467.3 (China)	Ensures the enclosure acts as a sufficient thermal barrier.
<i>Internal Thermal Runaway</i>	5 minutes delay of smoke to cabin	GB 38031 (China)	Provides occupants sufficient time to evacuate.
<i>Water & Dust Sealing</i>	IP67 / IP6K9K equivalent	Industry Standard	Prevents internal short circuits.
<i>Structural Frame Weight</i>	65 kg (Total Pack 543 kg)	Mass-Market Target	Maintains vehicle range efficiency.

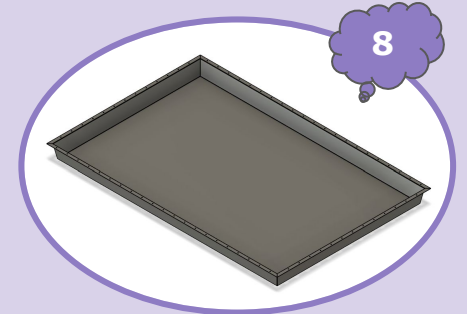
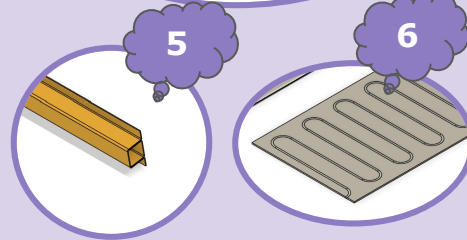
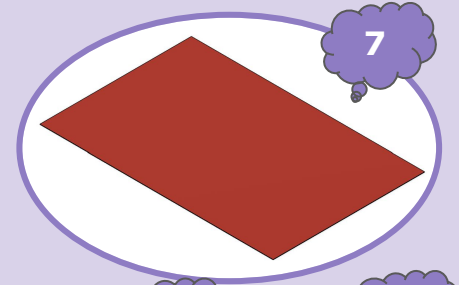
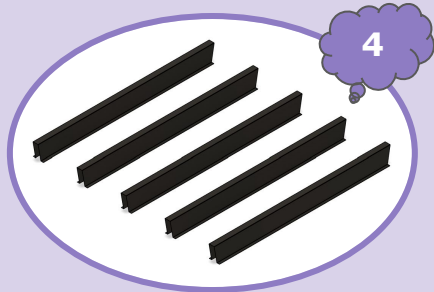
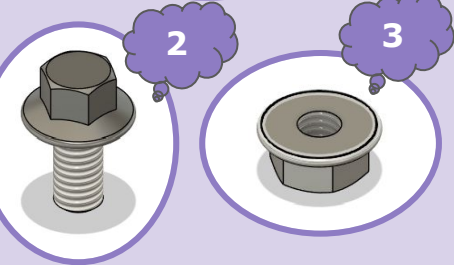
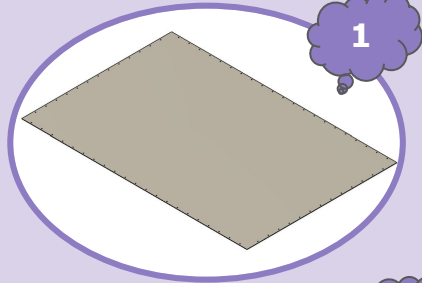
Project Focus

Not Considered

Exploded View



Item	Qty	Part Name	Material
1	1	Top Cover [2.5 mm]	CR5
2	68	Hex-Flange Bolts [ISO 4162 M6 x 12]	Steel 10.9
3	68	Hex-Flange Nuts [ISO 4161 M6]	Steel 10
4	5	Cross-Member Beams [1.0 mm]	MartINsite 1500
5	2	Side Rails [2.5 mm]	MartINsite 1500
6	6	Thermal Plates	Al 6061
7	1	Bottom Plate [1.5 mm]	MartINsite 1500
8	1	Main Housing Tray [1.5 mm]	Fortiform 1180



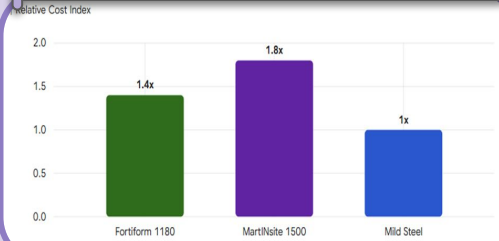
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Roll Forming - Side Rail (AHSS)



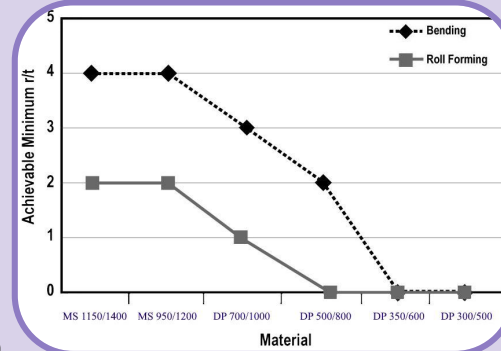
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Relative Cost Comparison



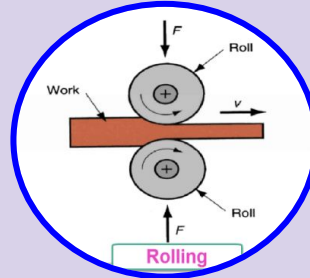
Material Property Comparison

Material	Yield Strength	Elongation
Fortiform 1180	1060	13%
MartInsite 1500	1500	3%

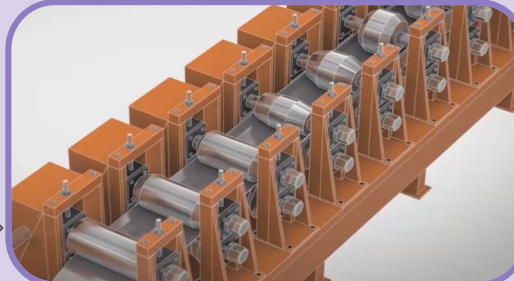


Roll forming is the optimal cold-forming process for AHSS rails:

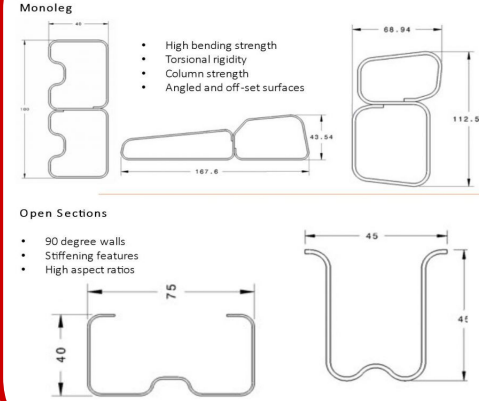
- **Incremental Deformation:** Gradually shapes metal through sequential passes to prevent cracking.
- **Springback Control:** Constant overbend distribution compensates for high residual stresses.
- **Profile Complexity:** Capable of producing complex open and closed profiles in high volumes.



For AHSS over 1000 MPa, require an R/t ratio of **3t to 4t**.



Roll Formed Profiles



Experimental Study on Steel ($Y > 1400$ MPa): Max load on each pass of 10-station rolling machine is **65kN**

shape

© 2023 Shape

Roll Forming - AHSS Guidelines

Part	Material	Corrosion Protection	Weldment	Justification
Cross-Members & Side Railing	MartINsite 1500	EG (Zn)	Resistance Spot Weldment	MartINsite microstructure alters upon heat treatment (HDGA). Current pulsation [3] during welding limits property degradation.

- **Resistance spot welding** [1] with complex stack-ups needs a complex, multi-pulse weld cycle.
- For steel-to-steel weld, coating must be removed.
- Coating displaces during welding at the peripheries; press-hardened Al-Si coating is harder to displace than cold stamped Zn coating.
- **Liquid Metal Embrittlement** [2] is the development of surface cracks during welding.

Figure 1

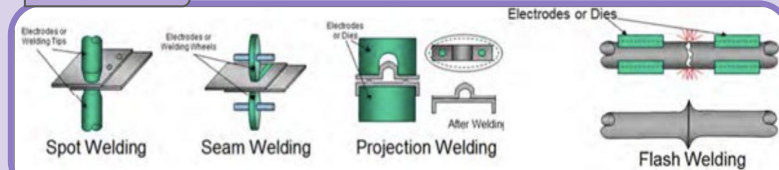


Figure 2

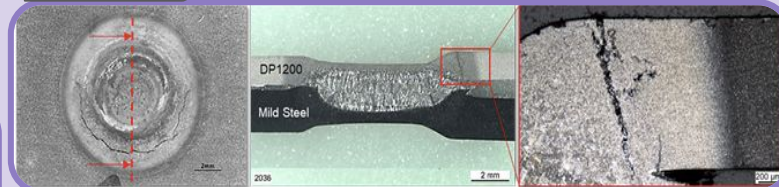
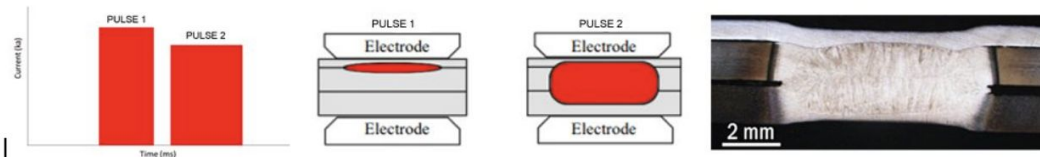
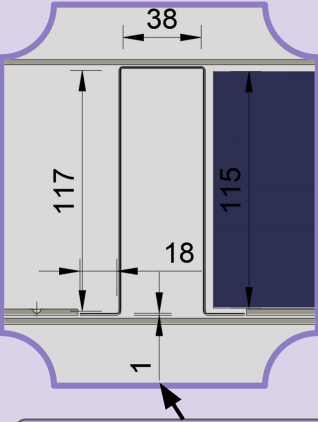


Figure 3

RSW Schedule for Joining 3T Stack-Up Using 2 Current Pulses



Considerations When Deciding Whether to Cold Form or Hot Form - AHSS Guidelines



Moment of Inertia

Part	I [mm ⁴]
CM	557k
SR	685k

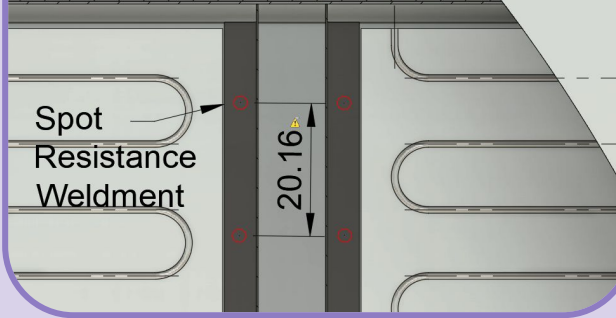
Cross-Member Side View



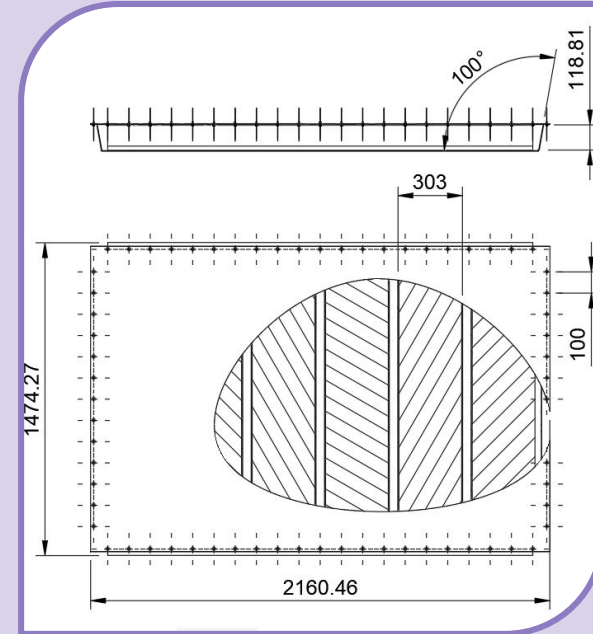
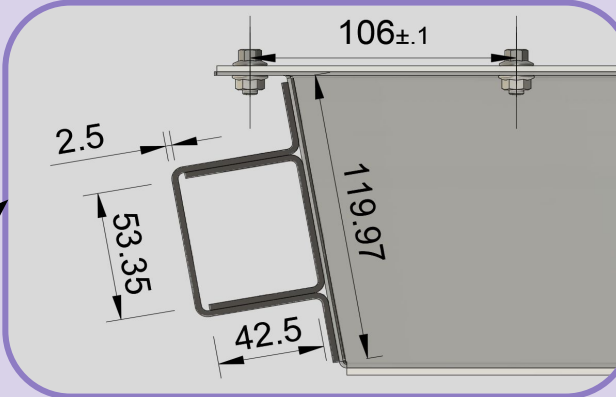
HexFlange Nut & Bolt

Side Rail Dimensions

Cross-Member Top View



BreakOut View

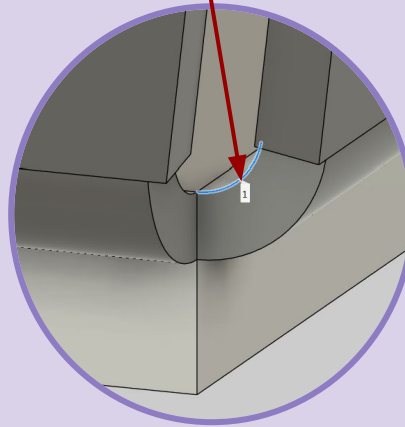


Tray Dimensions

Die Angle Requirements Table

Parameter	Value
Die Radius (R_i)	1.5 mm
Target Bend Angle ($\theta_{\square}$)	80 deg
Springback Factor ($K_{\square}$)	0.984
Final Part Radius (R)	1.52 mm
Required Die Angle (θ)	81.28 deg

Inner Bend Radius of Tray [4.5mm]

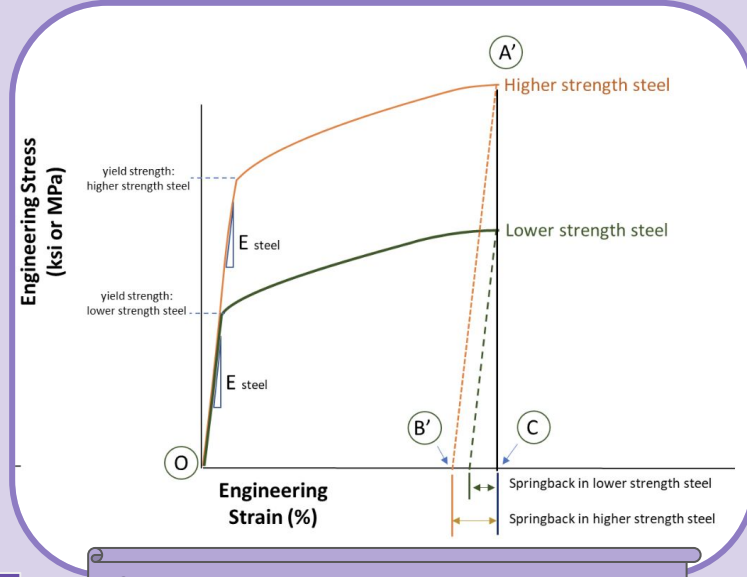


Springback - AHSS Guidelines

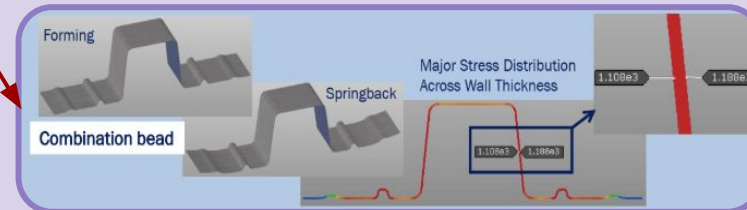
Draw & Stake bead for springback control during forming

Springback formula

$$K_s = 1 - 3 \left(\frac{Y \cdot R_i}{E \cdot t} \right) + 4 \left(\frac{Y \cdot R_i}{E \cdot t} \right)^3$$



Springback increases exponentially with material yield strength.





Beam Profile	Centroid (mm)	Moment of Inertia (mm ⁴)	Critical Buckling Stress (MPa)	Max Dynamic Stress (MPa)
Crossmember	60	557,065	7810	9571
Side Rail	29.7	684,969	1279	3490

Select Bolt Type

METRIC_Bolts

SELECT BOLT SIZE

M6

SELECT BOLT MATERIAL

Stainless Steel

Bolt Grip Length (mm)

5

ENTER TORQUE (Nm)

16

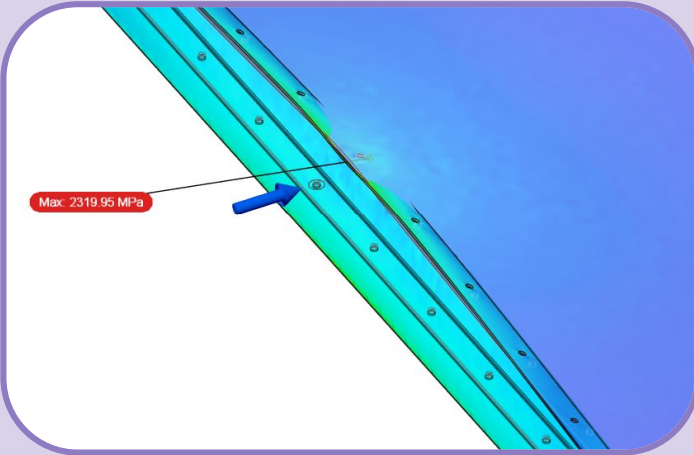
SELECT Lubricant / Bolt Condition:

Zinc Plated Steel (0.18)

GRIP LENGTH

CALCULATED OUTPUT:

Bolt Preload (N)	14815
Bolt Stress (Mpa)	828
Bolt Stretch (mm)	0.526
L/D Ratio	0.8



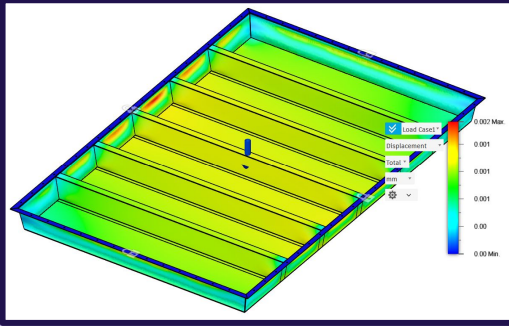
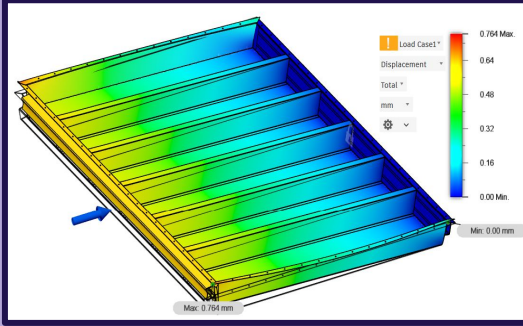
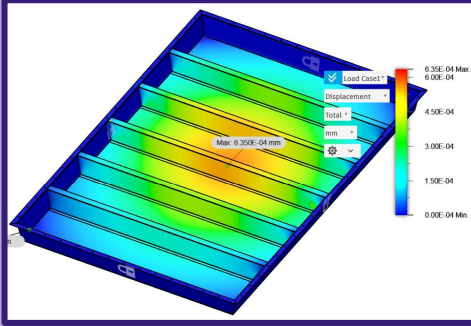
- Maximum stress on singular bolt & nut joint is 2320 MPa when 10k N side force is applied evenly.
- The nominal preload is ~ 81% of yield strength of material.
- Stress is 737 MPa of 900 MPa (material yield). Yield limit is 18k N for joint.

- Maximum Preload: $F_{PL,max} = (1 + 0.25) \cdot 14.8\text{kN} = 18.5\text{kN}$.
- Minimum Preload: $F_{PL,min} = (1 - 0.25 - 0.10) \cdot 14.8\text{kN} = 9.6\text{kN}$.

$$F_p = \frac{T}{K \cdot d} = 14.8\text{kN}$$

Centric load distribution

$$F_{shear} = \frac{P}{n}$$

Parameter	3G Evenly Distributed Top Load	100 kN Evenly Distributed Side Load	150 mm Bottom Intrusion (19,620 N)
Displacement Profile			
Von Mises Stress	Max ~ 7.43 MPa	Max ~ 817 MPa	Max ~ 8.68 MPa
Safety Factor	> 15	Min ~ 1.5	> 15

1. Chen, H., Zhao, L., Lu, S., Lin, Z., Wen, T., & Chen, Z. (2022). Progress and Perspective of Ultra-High-Strength Martensitic Steels for Automobile. *Metals*, 12(12), 2184. <https://doi.org/10.3390/met12122184>
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